Programski prevodioci

03 miniC i micko

25-26

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miniC i micko

- programski jezik miniC razvijen za potrebe nastave na FTN
- kompajler za jezik je: micko
- autori: doc dr Zorica Suvajdžin Rakić, prof dr Miroslav Hajduković, prof dr Žarko Živanov

- https://github.com/zsuvajdzin/micko
- zbirke (canvas)

U nastavku par napomena.

```
//OPIS: ABS funkcija
//RETURN: 5

int abs(int i) {
   int res;

   if(i < 0)
      res = 0 - i;
   else
      res = i;

   return res;
}

int main() {
   return abs(-5);
}
```

tokeni:

ID,

INT_NUMBER, UINT_NUMBER,

LPAREN, RPAREN, LBRACKET, RBRACKET,

SEMICOLON,

AROP, RELOP

ASSIGN,

TYPE,

IF, ELSE

RETURN

skip:

beline i linijski komentari

```
[ \t\n]+              { /* skip */ }

"int"                 { return _TYPE; }
"unsigned"            { return _TYPE; }
"if"                  { return _IF; }
"else"                { return _ELSE; }
"return"              { return _RETURN; }

"("                   { return _LPAREN; }
")"                   { return _RPAREN; }
"{"                   { return _LBRACKET; }
"}"                   { return _RBRACKET; }
";"                   { return _SEMICOLON; }
"="                   { return _ASSIGN; }

"+"                   { return _AROP; }
"-"                   { return _AROP; }

"<"                   { return _RELOP; }
">"                   { return _RELOP; }
"<="                  { return _RELOP; }
">="                  { return _RELOP; }
"=="                  { return _RELOP; }
"!="                  { return _RELOP; }

[a-zA-Z][a-zA-Z0-9]*  { return _ID; }
[+-]?[0-9]{1,10}      { return _INT_NUMBER;}
[0-9]{1,10}[uU]       { return _UINT_NUMBER;}

\/\/.*                { /* skip */ }
```

pojmovi, neterminali

program, function_list, function

type, parameter,

body, variable_list, variable,

statement_list, statement,

compound_statement, assignment_statement,

num_exp, exp, literal,

function_call, argument,

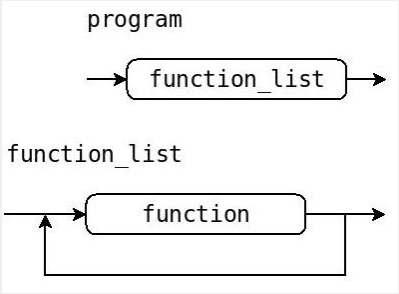
if_statement, if_part, rel_exp,

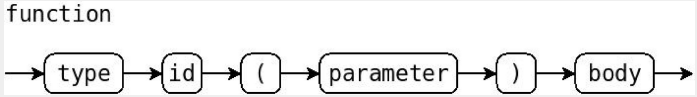
return_statement

```
%{
  #include <stdio.h>
  #include "defs.h"

  int yyparse(void);
  int yylex(void);
  ...
%}

%token _TYPE
%token _IF
...


%%

program
  : function_list
  ;

function_list
  : function
  | function_list function
  ;

function
  : type _ID _LPAREN parameter _RPAREN body
  ;

type
  : _TYPE
  ;

parameter
  : /* empty */
  | type _ID
  ;

body
  : _LBRACKET variable_list statement_list _RBRACKET
  ;

variable_list
  : /* empty */
  | variable_list variable
  ;

variable
  : _TYPE _ID _SEMICOLON
  ;
```

```
program
    : function_list
    ;

function_list
   : function
   | function_list function
   ;

function
   : type _ID _LPAREN parameter _RPAREN body
   ;
```

sintaksni dijagrami





Program ima bar jednu funkciju.

Obratiti pažnju:
- nigde ne piše da je to funkcija main().

```
type
  : _TYPE
  ;

parameter
  : /* empty */
  | type _ID
;

body
  : _LBRACKET variable_list statement_list _RBRACKET
  ;
```

```
✔ int f1(int p) {
    ...
  }

✔ int f2() {
    int x;
    int y;
    ...
  }

✖ int f(int p, int q) {
    ...
  }
```

Obratiti pažnju:

- nema liste parametara. Funkcija ima 0 ili 1 parametar.
- nema void. Funkcija mora vratiti int ili uint.
- u telu prvo idu deklaracije, a onda iskazi.

```
variable_list
  : /* empty */
  | variable_list variable
;

variable
  : _TYPE _ID _SEMICOLON
;
```

```
✔ int counter;
✔ int line;
✔ unsigned n;

✖ int counter, line;
```

Obratiti pažnju:

- nema višestruke deklaracije

```
statement_list
  : /* empty */
  | statement_list statement
  ;

statement
  : compound_statement
  | assignment_statement
  | if_statement
  | return_statement
  ;
```

Obratiti pažnju:

- compound_statement je vrsta statement-a

```
compound_statement
  : _LBRACKET statement_list _RBRACKET
  ;

assignment_statement
  : _ID _ASSIGN num_exp _SEMICOLON
  ;

return_statement
  : _RETURN num_exp _SEMICOLON
  ;
```

```
✔ { }
✔ { a = 8; }
✔ {
    if(a == b)
      counter = 0;
    a = 10;
  }
✖ {
    int a;
    a = 8;
  }
✖ return;
```

Obratiti pažnju:

- u bloku (složena naredba) nije dozvoljena deklaracija. Sve deklaracije moraju biti na početku funkcije.
- ne postoji void.

```
%nonassoc ONLY_IF
%nonassoc _ELSE

%%


if_statement
  : if_part %prec ONLY_IF
  | if_part _ELSE statement
  ;

if_part
  : _IF _LPAREN rel_exp _RPAREN statement
  ;
```

```
num_exp
  : exp
  | num_exp _AROP exp
  ;

exp
  : literal
  | _ID
  | function_call
  | _LPAREN num_exp _RPAREN
  ;

literal
  : _INT_NUMBER
  | _UINT_NUMBER
  ;
```

```
✔ a = b;
✔ counter = 0;
✔ line = line + 1;
✔ c = j + (k – 4);
✔ m = f();
✖ func();
✖ b = m * n;
✖ i = j++;
```

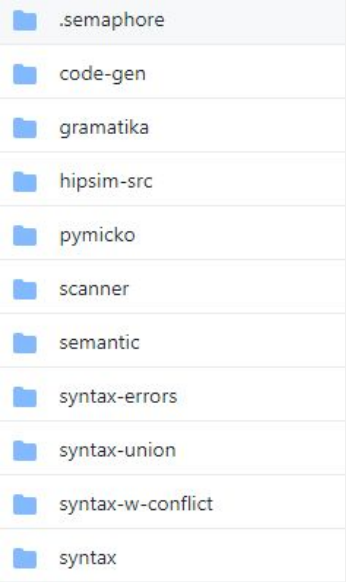
Podsetnik: num_exp se pojavljivao u assignment_statement i return_statement

```
function_call
  : _ID _LPAREN argument _RPAREN
  ;

argument
  : /* empty */
  | num_exp
  ;
```

Primetiti rekurziju koja nije direktna:
num_exp → function_call → argument → num_exp

```
rel_exp
  : num_exp _RELOP num_exp
  ;
```

Podsetnik: rel_exp se pojavljivao u if_part



- micko kompajler (GH)
- razdvojene su faze, poslednja code-gen